

Expectation Decider

Probability Insights Report — Student Competitive Exam Dataset

Dataset: 200 students | 5 features | Binary outcome: Pass / Fail

1. Dataset Overview

The dataset contains 200 student records with the following 5 features:

Feature	Type	Description
study_hours	Numeric	Weekly study hours
attendance	Numeric	Class attendance percentage
group_discussion	Categorical	Participation: Yes / No
previous_test_score	Numeric	Score in previous test
final_exam_pass	Categorical	Outcome: Pass / Fail

Pass: 142 students (71%) | Fail: 58 students (29%) | Total: 200

2. Basic Probability Events (Q1 & Q2)

Three Probability Events from the Dataset

Event	Calculation	Result
P(Pass)	$142 \div 200$	0.71 (71%)
P(Group Discussion = Yes)	$99 \div 200$	0.495 (49.5%)
P(Attendance > 75%)	$101 \div 200$	0.505 (50.5%)

Empirical vs Theoretical Probability

Type	Value	Meaning
Empirical P(Pass)	$142 \div 200 = 0.71$	Based on actual data
Theoretical P(Pass)	$1 \div 2 = 0.50$	Assuming equally likely outcomes

The empirical probability (0.71) is higher than theoretical (0.50), meaning real students pass at a rate well above chance — indicating that preparation and attendance do help.

3. Random Variable & Binomial Distribution (Q3)

X = Number of students passing out of $n = 3$ randomly selected students

$p = 0.71$ (from dataset) $n = 3$ $X \sim \text{Binomial}(3, 0.71)$

Binomial PMF Formula

$$P(X = k) = C(n, k) \times p^k \times (1 - p)^{(n - k)}$$

Probability Distribution Table

X (No. of Passes)	Formula	P(X)
$X = 0$	$C(3,0) \times 0.71^0 \times 0.29^3$	0.0244
$X = 1$	$C(3,1) \times 0.71^1 \times 0.29^2$	0.1791
$X = 2$	$C(3,2) \times 0.71^2 \times 0.29^1$	0.4386
$X = 3$	$C(3,3) \times 0.71^3 \times 0.29^0$	0.3579

Mean and Variance

$$\text{Mean } (\mu) = n \times p = 3 \times 0.71 = 2.13$$

$$\text{Variance } (\sigma^2) = n \times p \times (1 - p) = 3 \times 0.71 \times 0.29 = 0.6177$$

On average, 2.13 out of 3 randomly selected students are expected to pass.

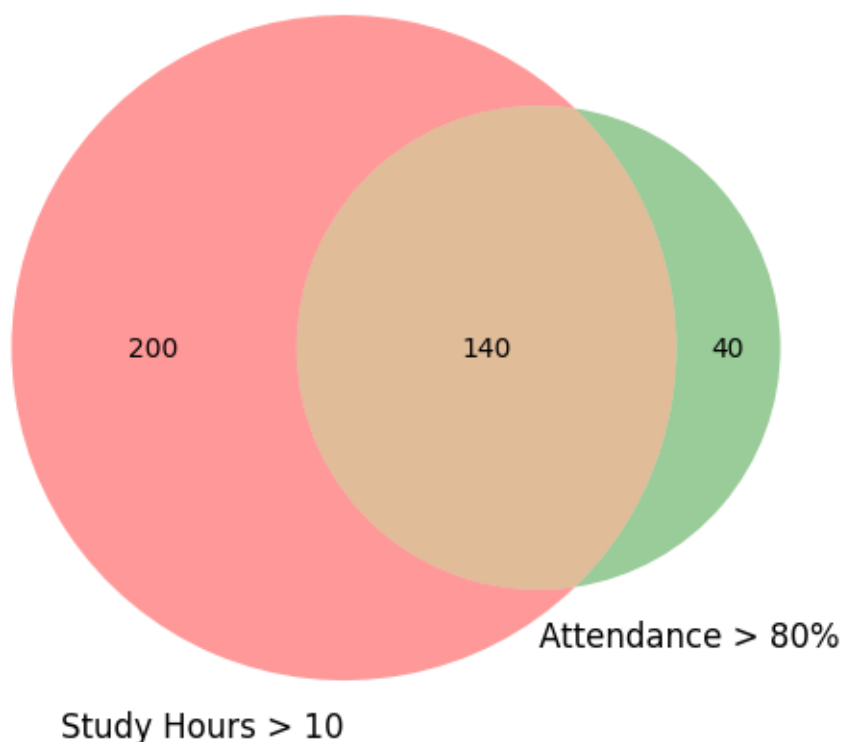
4. Venn Diagram — Study Hours & Attendance (Q4)

Sets defined: A = Study Hours > 10/week B = Attendance > 80%

Region	Count	Meaning
A only (Study > 10, Attendance $\leq$ 80%)	92	Hard workers, low attendance
$A \cap B$ (Both conditions met)	60	Disciplined students
B only (Attendance > 80%, Study $\leq$ 10)	18	Consistent attenders, low study hours
Neither	30	Low study hours & low attendance

60 students satisfy both conditions. The overlap ($A \cap B$) represents the most disciplined segment, who are likely strongest pass candidates.

Venn Diagram of Student Groups



5. Contingency Table & Probability (Q5)

Contingency Table: Group Discussion vs. Exam Result

Group Discussion	Fail	Pass	Total
No	35	66	101
Yes	23	76	99
Total	58	142	200

Probability Calculations

$$\text{Joint } P(\text{GD=Yes AND Pass}) = 76 \div 200 = 0.38$$

$$\text{Marginal } P(\text{Pass}) = 142 \div 200 = 0.71$$

$$\text{Conditional } P(\text{Pass} \mid \text{GD=Yes}) = 76 \div 99 = 0.7677$$

Students who joined group discussions pass at 76.77%, slightly higher than the overall 71%. The events are dependent (not independent) and not mutually exclusive.

6. Bayes' Theorem — Attendance & Passing (Q7)

Given Information

Symbol	Value	Meaning
$P(H \text{Pass})$	0.70	70% of passers had high attendance
$P(H \text{Fail})$	0.40	40% of failers had high attendance
$P(H)$	0.60	60% of all students had high attendance
$P(\text{Pass})$	$142 \div 200 = 0.71$	Overall pass rate from dataset

Step-by-Step Calculation

$$\begin{aligned}
 P(\text{Pass} | H) &= [P(H | \text{Pass}) \times P(\text{Pass})] \div P(H) \\
 &= [0.70 \times 0.71] \div 0.60 \\
 &= 0.497 \div 0.60 \\
 &= 0.8283 \rightarrow 82.83\%
 \end{aligned}$$

A student with high attendance (> 80%) has an 82.83% probability of passing — the highest probability found in this entire analysis.

7. Final Summary — What Factors Most Affect Passing?

Rank	Factor	Insight	Pass Probability
1	High Attendance (> 80%)	Bayes' Theorem result	82.83%
2	Study Hours > 10 + Attendance > 80%	Both conditions met (60 students)	Strong indicator
3	Group Discussion = Yes	Conditional probability	76.77%
4	Overall Base Rate	Empirical probability	71.00%
5	Group Discussion = No	Among non-participants	65.35%

Key Takeaways:

- Attendance is the strongest predictor — students with >80% attendance have an 82.83% chance of passing.
- Group Discussion participation boosts pass probability from 71% (base) to 76.77%.
- 60 students combined high study hours AND high attendance — the ideal student profile.
- The gap between empirical (71%) and theoretical (50%) probability shows preparation genuinely matters.

- Expected passes in any group of 3 students = 2.13, confirming that passing is the more likely outcome overall.